Satcom Purchase Management System Documentation

1. Overview

The **Satcom Purchase Management System** is a robust, web-based platform designed to digitize and streamline the procurement lifecycle within the Satcom department. It enables users to create, manage, and track purchase proposals through multi-level approvals, document submissions, and order processing.

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2. Key Features

2.1 User Management

- Role-based access: Admin, Super Admin, Regular Users
- Secure authentication & authorization
- Custom permissions and access restrictions

2.2 Proposal Management

- Create and submit purchase proposals
- Attach relevant supporting documents
- Real-time proposal tracking and status updates
- Automated multi-stage approval workflow

2.3 Workflow Stages

The system includes the following detailed workflow stages:

1. **Intender Note In Approval** – Initial note from the intender
2. **Intender Note Approved**
3. **Agenda Put Up For SPC**
4. **Agenda Put Up For GPC**

5. **SPC Approved** – Approval from SPC committee
6. **GPC Approved** – Approval from GPC committee
7. **Bid Floated on GeM Portal** – Floating the bid document
8. **BID Extended**
9. **Bid Open**
10. **Technical Evolution In Progress** – Technical evaluation report
11. **Received Representation**
12. **Representation Reply Submitted On GeM**
13. **Financial Approval** – Financial approval process
14. **BID In Negotiation** – Price negotiation with vendors
15. **PO Placed** – Purchase order placement
16. **Item Delivered** – Item delivery confirmation
17. **Payment Completed**
18. **Bid External** – External bid process
19. **TEC Approved** – Technical evaluation committee approval
20. **Deadstock Entry** – Entry in deadstock register
21. **MII Remove Approval**
22. **Bid Cancle**
23. **TEC Report Submitted On GeM Portal**
24. **BID-Financial Opened**
25. **Archive**

2.4 Document Management

- Secure, structured file uploads
- Support for various formats: PDF, DOC, XLS, JPG, PNG
- Stage-wise document organization and version control

- Document tagging for quick search and retrieval

2.5 Notification System

- Email alerts for approvals and updates
- Assignment notifications to relevant users
- Upload and status change alerts

2.6 Reporting & Analytics

- Dashboard for proposal status overview
- Exportable reports for management
- Visualizations using Chart.js

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3. Technical Architecture

3.1 Frontend

- HTML5, CSS3, JavaScript
- Bootstrap framework
- SweetAlert2 for interactive modals and alerts
- Chart.js for analytics
- FontAwesome for iconography

3.2 Backend

- PHP 7.4+
- MySQL/MariaDB
- PDO for database interactions
- PHPMailer for email handling

3.3 Security Features

- Session-based login

- SQL injection protection using prepared statements
- Cross-site scripting (XSS) and CSRF prevention
- Validated file uploads

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4. Database Structure

Key Tables:

1. `users` – Stores user details and roles
2. `users_backup`, `users_backup_before_group_removal`, `users_backup_limited_role`, `users_role_backup` – User data snapshots
3. `user_groups` – User group definitions
4. `user_group_assignments` – Group-to-user mappings
5. `proposals` – Stores proposal data
6. `proposal_assignments` – Tracks user-task mapping
7. `proposal_documents` – Manages attached files
8. `proposal_followups` – Follow-up records for proposals
9. `proposal_group_assignments` – Group-to-proposal mapping
10. `proposal_changes` – Audit log of proposal modifications
11. `proposal_dynamic_fields` – Custom fields for proposals
12. `proposal_email_log` – Emails triggered per proposal
13. `proposal_history` – Historical state tracking
14. `proposal_stage_fields`, `proposal_stage_field_values`, `proposal_stage_field_value_history` – Stage-specific data
15. `proposal_stages` – Tracks stage transitions and approvals
16. `workflow_configurations`, `workflow_stage_fields` – Defines stage flows and configuration
17. `stage_fields`, `stage_field_configurations`, `stage_field_values` – Dynamic field settings
18. `activity_log`, `general_activity_log` – Logs for user/system activities

19. `background_email_queue` – Pending email tasks
20. `notifications` – In-app/email notifications
21. `followups` – Independent follow-up tracking

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5. Directory Structure

```
purchase/
├── api/                # API endpoint handlers
├── assets/             # Static resources (CSS, JS, images)
│   ├── css/
│   ├── js/
│   ├── fonts/
│   └── images/
├── config/             # Configuration files
├── uploads/            # Uploaded documents
│   ├── documents/
│   └── proposals/
└── vendor/             # Composer/vendor packages
```

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6. Installation Requirements

Server Requirements:

- Apache or Nginx
- PHP 7.4 or higher
- MySQL 5.7 or above
- `mod_rewrite` module enabled
- GD Library & FileInfo extension

Required PHP Extensions:

- PDO

- PDO MySQL Driver
- Mbstring
- XML
- JSON

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7. Setup Instructions

Step 1: Database Setup

Execute the following SQL files in order:

```
- setup_database.sql
- create_tables.sql
- workflow_configurations.sql
```

Step 2: Configuration

- Copy `config.php` to `config/`
- Update DB credentials, base URL, and email settings

Step 3: Permissions

```
chmod -R 755 uploads/
chmod -R 755 assets/
chmod 644 config/config.php
```

Step 4: Initial Application Setup

- Create the first Admin user
- Define workflow stages and assign roles
- Set up departments and email templates

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8. Security Considerations

8.1 File Upload

- Enforce file type restrictions and size limits
- Rename files securely to avoid path traversal
- Optionally integrate antivirus scanning

8.2 Access Control

- Role and permission-based system
- Optional IP restriction
- Strict session timeout policies

8.3 Data Protection

- Regular data backups
- HTTPS for secure communication
- Sensitive data encryption and sanitization

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9. Version History

Version	Date	Notes
v1.0.0	2025-07-01	Initial release with core features
